

Dynamics of Open Quantum Systems I: Density Operator and Fermi's Golden Rule

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This note reviews the density operator method to describe the time evolution of quantum systems. It bridges the dynamics of a single quantum system, which is described by the Schrödinger equation, and the dynamics of a thermal ensemble of systems, which is described by the master equation.

1 Density Operator and Density Matrix

In quantum mechanism, a complete description of the system can be achieved if we know the complete set of mutually commuting operators, so all corresponding physical quantities can be measured simultaneously. For hydrogen atom, this set of operators could be $\hat{H}$, $\hat{L}^2$, $\hat{L}_z$, $\hat{S}^2$ and $\hat{S}_z$. A full set of measurements produces a **pure state** $|\Psi\rangle$, which is the eigenstate of the commuting operators, such as the $|\phi_{1s}\alpha\rangle$ state of the H atom. The expectation value of the observable $\hat{O}$ is

$$\langle\hat{O}\rangle = \langle\Psi|\hat{O}|\Psi\rangle \quad (1)$$

There are occasions that the complete set of mutually commuting operators is unknown, or the full set of measurements has not been done. In such cases the state of this system will contain a statistical mixture of pure states, such state is known as a **mixed state**. Note that the mixed state is not a superposition of states, the later is still a pure state. Let $|\Psi_\mu\rangle$ be the pure states contained in a mixed state. The probability of finding $|\Psi_\mu\rangle$ in the mixed state is denoted as w_μ . Clearly, $\sum_\mu w_\mu = 1$. The expectation value of the observable $\hat{O}$ is thus

$$\langle\hat{O}\rangle = \sum_\mu w_\mu \langle\Psi_\mu|\hat{O}|\Psi_\mu\rangle \quad (2)$$

The **nonequilibrium statistical operator** or the **density operator** $\hat{W}$ for the mixed state is defined as

$$\hat{W} = \sum_\mu w_\mu |\Psi_\mu\rangle\langle\Psi_\mu| \quad (3)$$

One can shown that the expectation value of $\hat{O}$ can be written as

$$\langle\hat{O}\rangle = \text{Tr}(\hat{W}\hat{O}) \quad (4)$$

where Tr is the trace of the operator, $\text{Tr}\hat{A} = \sum_a \langle a|\hat{A}|a\rangle$ and $|a\rangle$ is a complete orthonormal basis set. Just like a pure state is described by a state vector $|\Psi\rangle$, a mixed state is described by the density operator.

The pure states $|\Psi_\mu\rangle$ contained in the mixed state can be time-dependent. The time evolution of $|\Psi_\mu\rangle$ follows the Schrödinger equation

$$i\hbar \frac{\partial}{\partial t} |\Psi_\mu(t)\rangle = \hat{H} |\Psi_\mu(t)\rangle \quad (5)$$

The formal solution of $|\Psi_\mu(t)\rangle$ can be expressed in terms of the time evolution operator $\hat{U}(t)$,

$$|\Psi_\mu(t)\rangle = \hat{U}(t) |\Psi_\mu(0)\rangle \quad (6)$$

We can then write the equation of motion for $\hat{U}(t)$,

$$i\hbar \frac{\partial}{\partial t} \hat{U}(t) = \hat{H} \hat{U}(t) \quad (7)$$

The time-dependent density operator can be written as

$$\begin{aligned} \hat{W}(t) &= \sum_{\mu} w_{\mu} |\Psi_{\mu}(t)\rangle \langle \Psi_{\mu}(t)| \\ &= \sum_{\mu} w_{\mu} \hat{U}(t) |\Psi_{\mu}(0)\rangle \langle \Psi_{\mu}(0)| \hat{U}^{\dagger}(t) \end{aligned} \quad (8)$$

Therefore,

$$\boxed{\hat{W}(t) = \hat{U}(t) \hat{W}(0) \hat{U}^{\dagger}(t)} \quad (9)$$

Using Eq. (7), one immediately have

$$\boxed{\frac{\partial}{\partial t} \hat{W}(t) = -\frac{i}{\hbar} [\hat{H}, \hat{W}(t)]} \quad (10)$$

where $[\hat{H}, \hat{W}(t)] = \hat{H}\hat{W}(t) - \hat{W}(t)\hat{H}$ is the commutator. This is the equation of motion of the density operator, also known as the **quantum Liouville equation** or the **Liouville-von Neumann equation**.

The density operator can also be represented using a complete orthonormal set of basis states $|a\rangle$,

$$\hat{W} = \sum_{a,b} |a\rangle \langle a| \hat{W} |b\rangle \langle b| \quad (11)$$

The elements $\rho_{ab} = \langle a| \hat{W} |b\rangle$ are called the **density matrix**. The diagonal element

$$\rho_{aa} = \sum_{\mu} w_{\mu} |\langle a| \Psi_{\mu} \rangle|^2 \quad (12)$$

gives the population of state $|a\rangle$ in the mixed state, while the off-diagonal elements ρ_{ab} describe the coherence of the states.

From Eq. (10) one can derive the equation of motion of the density matrix,

$$\frac{\partial}{\partial t} \rho_{ab}(t) = -\frac{i}{\hbar} \sum_c (H_{ac} \rho_{cb} - H_{cb} \rho_{ac}) \quad (13)$$

We can also separate the diagonal elements of the Hamiltonian, H_{aa} and H_{bb} , from the summation, Let

$$\omega_{ab} = \frac{H_{aa} - H_{bb}}{\hbar} \quad (14)$$

be the **transition frequency**, Eq. (13) becomes

$$\boxed{\frac{\partial}{\partial t} \rho_{ab}(t) = -i\omega_{ab}\rho_{ab} - \frac{i}{\hbar} \left(\sum_{c \neq a} H_{ac}\rho_{cb} - \sum_{c \neq b} H_{cb}\rho_{ac} \right)} \quad (15)$$

2 Generalized Master Equation

In this section we will study the dynamics of two coupled multi-level systems in a heat bath. Consider the transition from a continuous manifold of states $\{|a\rangle\}$ (the initial states) to another continuous manifold of states $\{|\alpha\rangle\}$ (the final states). A realistic example is the transition between one electronic state to another, and the continuous manifold can be the vibrational states. The initial and final states are coupled with each other through the matrix element $V_{a\alpha}$. We assume that coupling matrix element of the states within the initial or final manifolds are zero, i.e., $V_{aa'} = V_{\alpha\alpha'} = 0$. The Hamiltonian of the system can then be written as

$$\hat{H} = \sum_a E_a |a\rangle\langle a| + \sum_\alpha E_\alpha |\alpha\rangle\langle \alpha| + \sum_a \sum_\alpha (V_{a\alpha} |a\rangle\langle \alpha| + V_{\alpha a} |\alpha\rangle\langle a|) \quad (16)$$

We further assume that the coupling between the system and the heat bath is stronger than the interstate coupling. In other words, **the transition rate within the two manifold is much faster than the transition between initial and final states**. This assumption ensures that **the states within the manifolds remain thermal equilibrium during the dynamics**. Therefore, the distribution of the initial and final states are

$$f(E_a) = \frac{1}{Z_i} e^{-\beta E_a} \quad (17a)$$

$$f(E_\alpha) = \frac{1}{Z_f} e^{-\beta E_\alpha} \quad (17b)$$

We want to know the time evolution of the population in initial states and final states, P_i and P_f , with

$$P_i(t) = \sum_a P_a(t) = \sum_a \rho_{aa}(t) \quad (18a)$$

$$P_f(t) = \sum_\alpha P_\alpha(t) = \sum_\alpha \rho_{\alpha\alpha}(t) \quad (18b)$$

In addition, the thermal equilibrium within the two manifolds lead to

$$P_a(t) = f(E_a) P_i(t) \quad (19a)$$

$$P_\alpha(t) = f(E_\alpha) P_f(t) \quad (19b)$$

Using Eq. (15), let μ and ν index both a and α ,

$$\frac{\partial}{\partial t} P_\mu(t) = -\frac{i}{\hbar} \sum_{\kappa \neq \mu} (V_{\mu\kappa} \rho_{\kappa\mu}(t) - V_{\kappa\mu} \rho_{\mu\kappa}(t)) = -\frac{i}{\hbar} \sum_{\kappa} (V_{\mu\kappa} \rho_{\kappa\mu}(t) - V_{\kappa\mu} \rho_{\mu\kappa}(t)) \quad (20)$$

and

$$\begin{aligned} \frac{\partial}{\partial t} \rho_{\kappa\mu}(t) &= -i\omega_{\kappa\mu} \rho_{\kappa\mu}(t) - \frac{i}{\hbar} \left(\sum_{\lambda \neq \kappa} V_{\kappa\lambda} \rho_{\lambda\mu}(t) - \sum_{\lambda \neq \mu} V_{\lambda\mu} \rho_{\kappa\lambda}(t) \right) \\ &\approx -i\omega_{\kappa\mu} \rho_{\kappa\mu}(t) - \frac{i}{\hbar} V_{\kappa\mu} (\rho_{\mu\mu}(t) - \rho_{\kappa\kappa}(t)) \\ &= -i\omega_{\kappa\mu} \rho_{\kappa\mu}(t) - \frac{i}{\hbar} V_{\kappa\mu} (P_\mu(t) - P_\kappa(t)) \end{aligned} \quad (21)$$

In the last lines we ignored the off-diagonal density matrix. The resulted equation is a first-order ODE that has an analytical solution,

$$\rho_{\kappa\mu}(t) = -\frac{i}{\hbar} V_{\kappa\mu} \int_0^t dt' e^{-i\omega_{\kappa\mu}(t-t')} [P_\mu(t') - P_\kappa(t')] \quad (22)$$

Insert this solution into Eq. (20) yields

$$\frac{\partial}{\partial t} P_\mu(t) = -\frac{1}{\hbar^2} \sum_{\kappa} |V_{\kappa\mu}|^2 2\text{Re} \int_0^t d\tau e^{-i\omega_{\kappa\mu}\tau} [P_\mu(t-\tau) - P_\kappa(t-\tau)] \quad (23)$$

Finally, setting $\mu = a$ and $\kappa = \alpha$, we get

$$\begin{aligned} \frac{\partial}{\partial t} P_i(t) &= \frac{\partial}{\partial t} \sum_a P_a(t) \\ &= -\frac{1}{\hbar^2} \sum_{a,\alpha} |V_{a\alpha}|^2 \int_0^t d\tau 2\text{Re} e^{-i\omega_{a\alpha}\tau} [P_a(t-\tau) - P_\alpha(t-\tau)] \\ &= -\frac{1}{\hbar^2} \sum_{a,\alpha} |V_{a\alpha}|^2 \int_0^t d\tau 2\text{Re} e^{-i\omega_{a\alpha}\tau} [f(E_a) P_i(t-\tau) - f(E_\alpha) P_f(t-\tau)] \end{aligned} \quad (24)$$

If we define the **memory kernel** as

$$K_{if}(\tau) = \frac{2}{\hbar^2} \sum_{a,\alpha} |V_{a\alpha}|^2 f(E_a) \cos(\omega_{a\alpha}\tau) \quad (25a)$$

$$K_{fi}(\tau) = \frac{2}{\hbar^2} \sum_{a,\alpha} |V_{a\alpha}|^2 f(E_\alpha) \cos(\omega_{a\alpha}\tau) \quad (25b)$$

the equation of motion of P_i and P_f become

$$\frac{\partial}{\partial t} P_i(t) = - \int_0^t d\tau [K_{if}(\tau) P_i(t-\tau) - K_{fi}(\tau) P_f(t-\tau)] \quad (26a)$$

$$\frac{\partial}{\partial t} P_f(t) = - \int_0^t d\tau [K_{fi}(\tau) P_f(t-\tau) - K_{if}(\tau) P_i(t-\tau)] \quad (26b)$$

These equations are called the **generalized master equation**. K_{if} and K_{fi} describes the transition rate between the initial and final states. The integral with respect to τ indicates that **the dynamics of the system at time t depends on its past states of all time before t .**

3 Pauli Master Equation and Fermi's Golden Rule

The generalized master equation describes a non-Markovian dynamics of the system. To further simplify the equation, here we will study the behavior of the memory kernel. The memory kernel K_{if} can also be written as (note that $\text{Re } e^{i\omega_{a\alpha}\tau} = \cos(\omega_{a\alpha}\tau) = \text{Re } e^{-i\omega_{a\alpha}\tau}$)

$$\begin{aligned} K_{if}(t) &= \frac{2}{\hbar^2} \text{Re} \sum_{a,\alpha} |\langle a|V|\alpha \rangle|^2 f(E_a) \exp\left(it \frac{E_a - E_\alpha}{\hbar}\right) \\ &= \frac{2}{\hbar^2} \text{Re} \sum_a \sum_\alpha f(E_a) \langle a|e^{\frac{iE_a t}{\hbar}} V e^{-\frac{iE_\alpha t}{\hbar}} |\alpha \rangle \langle \alpha|V|a \rangle \end{aligned} \quad (27)$$

Let $\hat{H}_0 = \sum_a E_a |a\rangle \langle a| + \sum_\alpha E_\alpha |\alpha\rangle \langle \alpha|$,

$$\begin{aligned} K_{if}(t) &= \frac{2}{\hbar^2} \text{Re} \sum_a \sum_\alpha f(E_a) \langle a|e^{\frac{i\hat{H}_0 t}{\hbar}} V e^{-\frac{i\hat{H}_0 t}{\hbar}} |\alpha \rangle \langle \alpha|V|a \rangle \\ &= \frac{2}{\hbar^2} \text{Re} \sum_a \langle a|f(E_a) e^{\frac{i\hat{H}_0 t}{\hbar}} V e^{-\frac{i\hat{H}_0 t}{\hbar}} V|a \rangle \\ &= \frac{2}{\hbar^2} \text{Re} \sum_a \langle a|\hat{W}_{a,\text{eq}} e^{\frac{i\hat{H}_0 t}{\hbar}} V e^{-\frac{i\hat{H}_0 t}{\hbar}} V|a \rangle \\ &= 2 \text{Re } C(t) \end{aligned} \quad (28)$$

where

$$C(t) = \frac{1}{\hbar^2} \text{Tr}_a(\hat{W}_{a,\text{eq}} V(t) V(0)) \quad (29)$$

is the **autocorrelation function of the interstate coupling** V . The equilibrium distribution $f(E_a)$ is replaced by the equilibrium density operator $\hat{W}_{a,\text{eq}}$. **The memory kernel is thus proportional to the autocorrelation function of the interstate coupling, which decays (exponentially) with time.**

For certain systems $C(t)$ decays very fast, therefore $K_{if}(\tau)$ is nonzero only when $\tau \rightarrow 0$. At this limit, $P_i(t - \tau) \approx P_i(t)$ and

$$\frac{\partial}{\partial t} P_i(t) = -k_{if} P_i(t) + k_{fi} P_f(t) \quad (30a)$$

$$\frac{\partial}{\partial t} P_f(t) = -k_{fi} P_f(t) + k_{if} P_i(t) \quad (30b)$$

This is the **Pauli master equation**. The transition rates k_{if} is defined as

$$k_{if} = \int_0^\infty d\tau K_{if}(\tau) = \frac{2}{\hbar^2} \sum_{a,\alpha} |V_{a\alpha}|^2 f(E_a) \text{Re} \int_0^\infty d\tau e^{i\omega_{a\alpha}\tau} \quad (31)$$

Here we can change the integration limit from t to ∞ also because $K_{if}(\tau)$ is nonzero only when $\tau \rightarrow 0$. Using the Fourier transform of the delta function,

$$\delta(\omega) = \frac{1}{2\pi} \int_{-\infty}^\infty dt e^{i\omega t} \quad (32)$$

we get

$$k_{if} = \frac{2\pi}{\hbar} \sum_{a,\alpha} |V_{a\alpha}|^2 f(E_a) \delta(E_a - E_\alpha) \quad (33)$$

This is the transition rate expression of **Fermi's golden rule**. As stated in the beginning, using density operator we connected the coherent dynamics (Schrödinger equation) and the incoherent dynamics (master equation and golden rule).

For the case that the initial and final states corresponds to the donor and acceptor electronic states and the manifolds $\{|a\rangle\}$ and $\{|\alpha\rangle\}$ corresponds to heat bath states of the polar solvent, Eq. (16) gives the Hamiltonian of the electron transfer system. Manipulation of the golden rule formula Eq. (33) will lead to the Marcus theory expression of the electron transfer rate.